

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Winter Examination-2023 Course: B. Tech. Branch: ALL Semester: III Subject Code & Name: BTBS301/ BTES 301 Engineering Mathematics-III Max Marks: 60 Date: 02.01.2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Find the Laplace transform of $f(t) = t^2 \sin 2t$	Understand/ CO1	6
B)	Find Laplace transform of $F(t) = \int_0^t \frac{e^{-at} - e^{-bt}}{t} dt$	Understand /CO1	6
C)	Find the Laplace transforms of $f(t) = \frac{t}{T}$, for $0 < t < T$ (saw - tooth wave function of period T)	Apply/CO1	6
Q.2	Solve Any Two of the following.		12
A)	Find inverse Laplace transform of $\cot^{-1}\left(\frac{s+3}{2}\right)$	Understand /CO2	6
B)	By using Partial fraction expansion to find inverse Laplace transform of $F(s) = \frac{s}{(s^2+1)(s^2+4)}$	Understand /CO2	6
C)	Using the Laplace transform, solve the differential equation $\frac{d^2x}{dt^2} + 9x = \cos 2t$; if $x(0) = 1$, $x\left(\frac{\pi}{2}\right) = -1$.	Apply/CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Express the function $f(x) = \begin{cases} 1 & \text{for } x \leq 1 \\ 0 & \text{for } x > 1 \end{cases}$ as a Fourier integral.	Understand /CO3	6
B)	Find the Fourier sine transform of $f(x) = e^{- x }$, and hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}$, $m > 0$.	Understand /CO3	6
C)	Using Parseval's identity, show that $\int_0^\infty \frac{t^2}{(4+t^2)(9+t^2)} dt = \frac{\pi}{10}$	Apply/CO3	6
Q.4	Solve Any Two of the following.		12
A)	Solve the following partial differential equations $(mx - ny)p + (nx - lz)q = ly - mx$	Understand /CO4	6

B)	A string is stretched and fastened to two points l apart. Motion is started by replacing the string in the form $y = A \sin \frac{\pi x}{l}$ from which it is released at time $t = 0$. Show that the displacement of a point at a distance x from one end at time t is given by $y(x, t) = A \sin \frac{\pi x}{l} \cos \frac{\pi ct}{l}$.	Apply/CO4	6
C)	Solve the following equation by the method of separation of variables: $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$, given that $u = 0$ when $t = 0$ and $\frac{\partial u}{\partial t} = 0$ when $x = 0$.	Apply /CO4	6
Q. 5	Solve Any Two of the following.		12
A)	If $f(z)$ is analytic, show that $\left[\frac{\partial f(z) }{\partial x}\right]^2 + \left[\frac{\partial f(z) }{\partial y}\right]^2 = f'(z) ^2$.	Understand /CO5	6
B)	Apply Cauchy's integral Formula to evaluate $\oint_C \frac{e^{-z}}{z+1} dz$, where C is the circle (a) $ z = 2$ and (b) $ z = \frac{1}{2}$.	Apply/CO5	6
C)	State Cauchy's residue theorem and evaluate $\oint_C \tan z dz$, where C is the circle $ z = 2$.	Apply /CO5	6
*** End ***			